

Task 1

	Title / Name	Type	Year	Link	Why useful	Key method / content
1	Thermal Imaging and Radar for Remote Sleep Monitoring of Breathing and Apnea	Paper	2024	https://arxiv.org/abs/2407.11936	First direct comparison of radar vs thermal imaging for non-contact sleep monitoring; clinically relevant apnea classification; provides hardware-synchronized data collection framework and open-source code.	Compares radar and thermal modalities for apnea detection (thermal outperforms radar); proposes multimodal method to classify obstructive vs central apnea; uses envelope detection to extract breathing waveform and detect apnea events and breathing rate.
2	UCLA-VMG / NonContactApneaDetection	Code	2024	https://github.com/UCLA-VMG/NonContactApneaDetection	Open-source implementation of data collection, preprocessing, and inference for radar + thermal sleep monitoring; reusable for Task 2 (envelope detection baseline) and Task 3 (data loading and preprocessing reference). Note: paper reports a dataset of 10 sleeping patients, but the repository does not include a data/ folder or download link — dataset availability is unconfirmed.	C++ data collection framework (Syndicate of Sensors, Boost-dependent); Python mmProc library (Python 3.8) for multimodal data processing; edf_proc for PSG ground truth; apnea_detection.py and breathing_detection.py for envelope-detection-based breathing and apnea estimation.
3	mmVital - Radar-based vital signs monitoring	Code	2024	https://github.com/UBiWeb-mmVital	Suite of AI/ML and signal processing techniques for non-contact vital signs	Repo contains: BeamSteering_PhaseCapture, MATLAB2Python, Radar2Jetson, TimeSeriesBasedDLforHR (12

	system			lab/m mVital	monitoring using mmWave radar. Designed for continuous HR/RR measurement of semi-stationary targets with various body positions and postures under dynamic and complex environments. Includes TI mmWave radar configurations, signal processing and HR/RR extraction methods, vibration simulation testbed, deep learning models, and mobile app for visualization.	architectures: BiLSTM, CNNLSTM, GRU, HARHN, Informer, LSTM, MTGNN, NBEATS, RNNED, Reformer, TPALSTM, Transformer), Vibration, VisualizeVital Signs, <code>vitalsign_rssd_code_1443</code> and <code>vitalsign_rssd_code_1843</code> .
4	A Real-Time Beam Steering and Accurate Vital Sign Estimation Method in an Indoor Environment	Paper	2024	https://doi.org/10.1109/JIOE.2024.3410209	Real-time beam steering for accurate vital sign estimation in indoor environments; directly relevant to multi-person monitoring and motion artifact mitigation. Code in <code>BeamSteering_Phase Capture</code> .	Beam steering + vital sign estimation; IEEE Internet of Things Journal.
5	Human Vital Signs Estimation Using Resonance Sparse Spectrum Decomposition	Paper	2024	https://doi.org/10.1109/THMS.2024.3381074	Introduces resonance sparse spectrum decomposition for vital signs estimation; useful for harmonic suppression and separating respiration/heartbeat components. Code in <code>vitalsign_rssd_code_1</code>	Resonance sparse spectrum decomposition (RSSD); IEEE Transactions on Human-Machine Systems.

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6	VS_DATASET (120 GHz radar vital signals)	Dataset	2024	https://github.com/Rc-W024/VSDATAS-ET	Two high-quality datasets collected with a 120 GHz FMCW Radar System-on-Chip (RSoC). Directly addresses the task's need for radar raw data + ground-truth vital signs. Covers both healthy and clinical cohorts, including apnea, making it relevant to respiration, heart rate, and apnea detection.	Healthy cohort: 24 subjects, Resting + Apnea (2-min each). Patient cohort: 15 participants, clinical data with specific anomalies and disease markers (10-min each). Collected using 120 GHz FMCW RSoC.
7	Example Code for mmWave Vital Signal Datasets	Code	2024	https://github.com/Rc-W024/VSDATAS-ET	Includes radar validation code (Test_speaker) covering the full pipeline from raw beat tone signal to target displacement signal extraction. Raw loudspeaker data from the test measurement is included, useful for Task 2 (phase → displacement) and Task 3 (preprocessing reference).	Radar validation code: raw beat tone signal → target displacement signal extraction. Includes raw loudspeaker test data. Recommends computing RR/HR on both radar and reference signals using the same method (IBI-based or spectrum-based) for fair comparison.
8	FMCW radar-based multi-person vital sign monitoring data	Dataset	2024	https://data.mendeley.com/datasets/684v4r8wfr	Directly addresses the task's need for public datasets with radar raw data + ground-truth vital signs, specifically for multi-person monitoring. Captures dual-subject respiration	Hardware: TI IWR6843ISK radar board + DCA1000EVM. Radar data in .bin format. Validation: ECG front-end, respiration and heartbeat waveforms in .csv format. Two scenarios: 1_AsymmetricalPosition, 2_SymmetricalPosition. Size: 1.5 GB.

				/1	and heartbeat using a 60 GHz FMCW radar, with synchronized ECG validation data.	Licence: CC BY 4.0.
9	Advances in Non-Contact Human Vital Sign Detection: A Detailed Survey of Radar and Wireless Solutions	Survey	2025	https://ieeexplore.ieee.org/document/10879370	Fills the survey gap (2023–2026 requirement). Open Access. Comprehensively covers radar and wireless technologies for non-contact vital sign detection, including FMCW, UWB, Doppler, stepped-frequency, and WiFi-based systems. Reviews signal processing pipelines, AI/ML methods, hardware designs, and application scenarios.	Covers: (1) FMCW radar systems; (2) UWB radar; (3) Doppler and stepped-frequency radar; (4) Other modalities; (5) AI techniques; (6) Signal processing chain: range-FFT → DC compensation → phase unwrapping → vibration FFT → range bin selection → vital rate estimation; (7) Challenges: motion artifacts, harmonics, clutter, multi-person separation, clinical validation.
10	A Novel Scheme of High-Precision Heart Rate Detection With a mm-Wave FMCW Radar	Paper	2023	https://doi.org/10.1109/ACCESS.2023.3303335	High-precision HR detection using mmWave FMCW radar; directly relevant to Task 2 (signal processing pipeline) and Task 3 (HR estimation). Reports MAE < 1 bpm. Open Access.	Adaptive range bin selection based on variance; smooth spline fitting for template signal; matched filtering to eliminate breathing and RBM interference; Variable Mode Extraction (VME) for heartbeat signal extraction; Double-Chirp Z-Transform (Double-CZT) for precise frequency measurement. MAE < 1 bpm. IEEE Access, Vol. 11, pp. 85118–85136, 2023.
11	Overview of mmWave Radar and Antenna Advances in	Survey	2025	https://doi.org/10.1109/RFM670	Complements the existing 2025 survey by focusing specifically on mmWave FMCW radar and high-performance	Reviews Doppler radar technologies for RVSM; antenna configurations for enhanced RVSM; comparative analysis of system architectures and antenna setups; challenges (respiratory

	Contactless Vital Signs Monitoring			34.2025.11284526	antenna designs (MIMO, leaky-wave) for remote vital signs monitoring (RVSM). Provides comparative analysis of system architectures and antenna configurations. Directly addresses challenges: respiratory harmonics, motion artifacts, and multi-target interference.	harmonics, motion artifacts, multi-target interference); future directions (adaptive signal processing, reconfigurable antennas, low-power architectures). Published in 2025 IEEE International RF and Microwave Conference (RFM), Penang, Malaysia.
12	State-of-the-Art Technologies and Emerging Trends in Radar-Based Vital Signs Sensing: A Review	Survey	2026	https://doi.org/10.1109/JSEN.2025.3646149	Most systematic and directly relevant review found so far. Covers the full chain from physiological basis to mathematical representation to signal processing algorithms to applications. Published in IEEE Sensors Journal (top venue). Includes both traditional frameworks and machine learning-driven approaches.	Sections: (I) Introduction; (II) Physiological Basis and Principles; (III) Signal Processing Algorithms (traditional + ML); (IV) Challenges and Future Trends; (V) Applications (bedside clinical monitoring, home healthcare, HCI, security surveillance, IoT). IEEE Sensors Journal, Vol. 26, Issue 3, pp. 3446–3459, Feb 2026.
13	Accurate multi-target vital signs detection method for FMCW radar	Paper	2023	https://doi.org/10.1016/j.measurement.2023.113715	Directly addresses multi-target vital signs detection, filling the gap in multi-person monitoring methods. Combines clutter suppression with multi-target range bin detection. Relevant to	Combines Moving Target Indication (MTI), Constant False Alarm Rate (CFAR) detection, K-means clustering, and local search with moving average filtering. Uses extended differentiate and cross multiply (DACM) algorithm for phase demodulation. Two-stage detection. Mean accuracies: 94.2% (breathing rate), 95.1% (heartbeat

					Task 2 (phase demodulation via DACM) and Task 3 (multi-target data processing).	rate). Published in Measurement, 2023.
14	VitalSign-Capstone-2023	Code	2023	https://github.com/s-hakhal350/VitalSign-Capstone-2023	Vital-sign-specific code repository using FMCW radar + deep learning for heart and breathing rate monitoring. Directly relevant to Task 3 (end-to-end AI pipeline) and useful for Task 2 as a comparison baseline. Includes irregularity detection and notification system.	Indoor radar-based vital sign monitoring system. Uses FMCW radar technology and deep learning to track heart and breathing rates. Detects irregularities and notifies a connected laptop. Capstone project, 2023.
15	MobiVital-dataset	Dataset	2025	https://github.com/n-esl/MobiVital-dataset ; Data: https://doi.org/10.5281/zenodo.15022885	24-hour large-scale dataset from 12 subjects using IR-UWB radar, with time-synchronized ground truth from wearable chest bands and finger clips. UCLA IRB approved (IRB#23-000754). Directly satisfies the task's need for public datasets containing radar raw data + ground-truth vital signs. Includes both tripod-mounted and handheld scenarios, plus IMU data for platform movement estimation.	Data format: .csv table, 1500 rows ($30\text{ s} \times 50\text{ Hz}$) $\times$ 254 columns. Columns 1–6: IMU (accel x/y/z, gyro x/y/z). Columns 13–252: UWB data (120 in-phase + 120 quadrature). Distance resolution: 0.0514 m/column, starting at 0.30 m. Column 253: ground truth respiration waveform. Column 254: ground truth blood pulse waveform. Processing: read_parsed_data.m (MATLAB). GitHub repo contains 8 sample .csv files + processing script; full dataset on Zenodo.

Summary

Current mmWave radar vital sign detection has formed a standard pipeline of "range-FFT → phase extraction → denoising/separation → frequency estimation," with deep learning rapidly permeating the field; however, motion artifacts, respiratory harmonics, multi-person separation, and clinical validation remain the four core challenges. If starting a project tomorrow, I would recommend using survey #12 as the entry point, VS_DATASET as the data foundation, and mmVital as the code reference.